

ECC Pipeline — Complete Field Reference

Every field in the records table, every possible value, and what it means.

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1. Input Fields

These are written once when the record is loaded from the input JSONL. They never change.

unique_id

Composite key that ties this record back to the original filing.

Format: {filing_id}__{agent_id}

Example: 142C24A7FA176718

business_name

Full legal business name from the NC Secretary of State filing.

Example: THE CAROLINA CIVIC CENTER FOUNDATION, INC.

agent_name

The registered agent or officer of record. Format is typically LAST, FIRST MIDDLE.

Example: SCEIFORD, RICHARD

state

Two-letter state abbreviation.

Example: NC

jurisdiction

Full state name as recorded in the filing.

Example: NORTH CAROLINA

position_type

The role this person holds in the entity. Drives whether the pipeline searches for a personal or generic email.

Value	Meaning	Strategy triggered
Agent	Registered agent	without
Officer	Corporate officer	with
Director	Board director	with
Manager	LLC manager	with
Member	LLC member	with
Partner	Partnership partner	with

name_entity_type

Whether the agent is a person or a corporate entity.

Value	Count (500-record run)	Meaning
Person	493	Individual — name parsing applies
Organization	7	Corporate registered agent (e.g. CT Corporation System) — org strategy always used

is_org_agent

Computed from name_entity_type. 1 if the agent is an organization, 0 if a person.

When 1, the pipeline uses the without strategy regardless of position_type.

2. Discovery Fields

These are written by the ProducerWorker during the discovery phase.

strategy

Which email search approach was used for this record.

Value	When assigned	What it changes
with	position_type is Officer/Director/Manager AND agent is a person AND name has first+last	Searches by agent name; generates personal patterns (first.last, flast, etc.); snippet emails filtered to name-matching only
without	position_type is Agent, or agent is an org, or name parsing fails	Searches by business name only; generates generic patterns (info, contact, hello, etc.)

candidate_domain

The confirmed business domain, if one was found.

Value	Meaning
carolinaciviccenter.com	DNS or Serper found a domain with a working mail server
NULL	Neither DNS nor Serper could confirm a domain for this business

discovery_source

How the domain or email was found.

Value	Meaning
dns	MX record resolved for a domain derived from the business name
serper	Serper (Google) returned either a domain or an email for this business
input	Email address was already present in the input record — no discovery needed
NULL	Record is DISCOVERY_FAILED and nothing was found

Note: Before the fix applied in this run, records where Serper found an email in a snippet but no domain was established would incorrectly show NULL. After the fix, these correctly show serper.

discovery_attempts

How many times the discovery phase ran for this record. Normally 1. Increments when a transient error causes the record to enter DISCOVERING state and be retried.

mx_provider

The MX hostname returned by DNS for the candidate domain. This field drives the validation strategy in the consumer.

Pattern	Provider	Consumer action
*.mail.protection.outlook.com	Microsoft 365 / Exchange Online	MS probe runs first (free); Zuhai is fallback
aspmx.l.google.com	Google Workspace	Zuhai only
smtp.secureserver.net	GoDaddy Email	Zuhai only
mx.zoho.com	Zoho Mail	Zuhai only
mx1.titan.email	Titan / Neo Email	Zuhai only
smtp.google.com	Google (consumer)	Zuhai only
NULL	No MX record found	MS probe skipped; Zuhai only if candidates exist

candidate_email

The first (highest-ranked) email candidate. This is the email the consumer tries first.

candidate_emails

JSON array of all email candidates generated for this record, in priority order.

Example: ["director@carolinaciviccenter.com", "info@carolinaciviccenter.com", "contact@carolinaciviccenter.com"]

The consumer works through this list in order. If the first candidate fails, it tries the second, and so on.

subdomain_emails

JSON array of emails found on subdomains of the candidate domain (e.g. mail.domain.com). Stored separately because subdomain emails are lower-confidence and not used as primary candidates unless all main candidates fail.

3. Validation Fields

These are written by the ConsumerWorker after validation runs.

zuhai_status

The raw verdict from whichever validation service confirmed or rejected the email.

Value	Source	Meaning
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valid	Zuhal	Mailbox confirmed to exist and accept mail — highest confidence
accept-all	Zuhal	The mail server accepts all addresses without rejection (catch-all domain); individual existence cannot be confirmed
ms_valid	MS probe	Microsoft GetCredentialType confirmed the Azure AD user exists — free, no Zuhal cost
bbops_valid	bbops.io	SMTP-level confirmation from the bbops batch verifier
unknown	Zuhal	SMTP server gave an inconclusive response: timeout, greylisting, or a soft rejection that does not definitively confirm or deny existence
invalid	Zuhal or MS probe	Address definitively does not exist — SMTP server rejected with a hard 550/551
disposable	Zuhal	Address is on a known throwaway / temporary email provider
NULL	—	Record never reached validation (DISCOVERY_FAILED)

zuhal_score

Additive confidence score computed at validation time. Integer 0 through 4.

Scoring rules:

Criterion	Points	When it applies
Email domain fuzzy-matches candidate domain (RapidFuzz ratio >= 85)	+1	Always
Local part fuzzy-matches agent name variants (ratio >= 75)	+1	with strategy only
Local part is NOT a generic prefix	+1	with strategy only
Verdict is valid (not accept-all)	+1	with strategy
Local part IS a generic prefix	+1	without strategy
Verdict is valid (not accept-all)	+1	without strategy

Generic prefixes: info, contact, hello, admin, support, sales, help.

Score thresholds:

Score	confidence_tier
3 or 4	high
2	medium
0 or 1	low

verdict

Mirrors zuhal_status but also captures intermediate verdicts (e.g. valid even when ms_valid was the method). Used internally for state machine transitions. In practice identical to zuhal_status for all current validation methods.

validation_attempts

How many candidate emails were tried before reaching a final verdict. 0 means the record never entered validation (DISCOVERY_FAILED).

4. State Machine Fields

record_state

The current lifecycle position of the record. This is the authoritative field — the CSV and JSON are derived from it.

Value	Set by	Next valid states	Terminal
RAW	init_db (default)	DISCOVERING, DISCOVERED, DISCOVERY_FAILED	No
DISCOVERING	Producer (transient error)	DISCOVERED, DISCOVERY_FAILED	No
DISCOVERED	Producer (domain or email found)	VALIDATING	No
VALIDATING	Consumer (atomic claim)	VALIDATED, VALIDATION_FAILED, COST_SKIPPED	No
VALIDATED	Consumer	—	Yes
VALIDATION_FAILED	Consumer	DISCOVERED (via reset)	Yes

DISCOVERY_FAILED	Producer	RAW (via reset)	Yes
COST_SKIPPED	Consumer	DISCOVERED (via reset + higher ceiling)	Yes

Terminal states can be re-queued manually via:

```
python -m pipeline reset --db output/.../pipeline.db --status discovery_failed
python -m pipeline reset --db output/.../pipeline.db --status validation_failed
```

5. Audit Fields

process_trace

JSON array written at each pipeline stage. Every entry records what happened and how long it took.

Entry format:

```
{
  "stage": "dns",
  "outcome": "hit",
  "ms": 2896,
  "domain": "carolinaciviccenter.com"
}, {
  "stage": "dns",
  "outcome": "miss",
  "ms": 6182,
  "domain": null
}, {
  "stage": "patterns",
  "outcome": "gen",
  "n": 3
}, {
  "stage": "serper",
  "outcome": "hit",
  "ms": 11551
}, {
  "stage": "serper",
  "outcome": "miss",
  "ms": 10040
}, {
  "stage": "ms_api",
  "outcome": "valid",
  "ms": 843,
  "email": "info@domain.com"
}, {
  "stage": "zuhal",
  "outcome": "valid",
  "ms": 13397,
  "email": "director@carolinaciviccenter.com"
}, {
  "stage": "zuhal",
  "outcome": "invalid",
  "ms": 8200,
  "email": "info@domain.com"
}, {
  "stage": "input",
  "outcome": "hit",
  "ms": 0
}
```

Stage values:

Stage	Written by	When
dns	Producer	After DNS probe completes (hit or miss)
patterns	Producer	After email patterns are generated
serper	Producer	After Serper enrichment call
input	Producer	When email was already in the input record
ms_api	Consumer	After MS GetCredentialType call
zuhal	Consumer	After each Zuhal validation attempt

created_at

UTC timestamp when the record was first inserted into the database.

updated_at

UTC timestamp of the last write to this row. Updated on every state transition, validation attempt, and process trace append.

6. Output CSV Fields

The valid_emails.csv contains only VALIDATED records. Nine columns:

Column	Source field	Possible values
unique_id	records.unique_id	{filing_id}__{agent_id}
business_name	records.business_name	Free text
agent_name	records.agent_name	Free text
state	records.state	NC, TX, FL, etc.
email	records.candidate_email	Validated email address
zupal_status	records.zupal_status	valid, accept-all, ms_valid, bbops_valid
confidence_tier	computed from zupal_score	high, medium, low
discovery_method	records.discovery_source	dns, serper, input
validation_method	derived from zupal_status	zupal, ms_probe, bbops

validation_method derivation:

zupal_status	validation_method
ms_valid	ms_probe
bbops_valid	bbops
valid	zupal
accept-all	zupal
anything else	unknown

Summary: field coverage by record state

Field	RAW	DISCOVERED	VALIDATED	DISCOVERY_FAILED
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unique_id	set	set	set	set
business_name	set	set	set	set
strategy	set	set	set	set
candidate_domain	—	set or NULL	set or NULL	NULL
discovery_source	—	set	set	NULL
mx_provider	—	set or NULL	set or NULL	NULL
candidate_emails	—	set	set	NULL
zuhal_status	—	—	set	NULL
zuhal_score	—	—	set	NULL
verdict	—	—	set	NULL
process_trace	—	partial	full	partial